

ARKO

c.arko.business@gmail.com | Remote, US | [linkedin.com/in/arkchakrabarti](https://www.linkedin.com/in/arkchakrabarti) | github.com/SA-Ark

TECHNICAL SKILLS

AI/ML: Production RAG Systems, Large Language Models (LLMs), AI Agents, Model Context Protocol (MCP), Vector Search (pgvector, Pinecone), LangChain/LangGraph, Prompt Engineering, AI Evaluation, Fine-tuning, Structured Output

Languages: Python, TypeScript, Rust, SQL

Stack: FastAPI, Next.js, React, PostgreSQL, pgvector, Redis, Whisper, Docker, AWS, Kubernetes, CI/CD

PROFESSIONAL EXPERIENCE

Independent AI & Software Consultant 2024 – Present

AI Integration Engineer — Production AI systems with evaluation pipelines and measurable ROI for enterprise clients

Meridian Health Partners (Healthcare, 500+ Employees) — 8 months

- Deployed production RAG pipeline processing 200K+ clinical documents with 94% retrieval accuracy and <2s latency, enabling AI-powered diagnostic support across 12 hospital departments
- Built HIPAA-compliant AI document summarization reducing physician review time from 45 minutes to 5 minutes per patient intake, processing 3,000+ documents daily
- Designed AI evaluation framework with automated hallucination detection, drift monitoring, and A/B testing, maintaining 99.2% factual accuracy across 50K+ weekly AI-generated summaries

Vaulted Financial (Series B Fintech, \$15M Raised) — 5 months

- Architected AI-powered fraud detection pipeline using ensemble ML models, blocking \$3.2M in fraudulent transactions in first quarter while reducing false positives by 73%
- Built conversational AI assistant handling 8,000+ daily customer queries with 89% resolution rate, deflecting 55% of Tier-1 support and saving \$240K annually

ClearView Media (Digital Agency, 50+ Brand Clients) — 4 months

- Engineered AI content generation platform producing 500+ pieces of optimized marketing copy weekly, reducing content production costs by 70% while maintaining brand consistency across 50 accounts
- Deployed AI analytics dashboard with natural language querying, enabling non-technical managers to generate client reports in 2 minutes vs. previous 3-hour manual process

AI Data Specialist — Scale AI 2023 (6 months)

- Evaluated large language models through expert human feedback on reasoning, code generation, and instruction following, directly improving model quality for production deployment
- Performed AI safety red-teaming identifying critical failure modes including hallucination patterns, prompt injection vulnerabilities, and bias in production LLM outputs

Full-Stack Software Engineer — Contract 2020 – 2023

- Shipped 8 production applications with AI integrations including intelligent search, recommendation systems, and automated content generation serving 500K+ monthly active users
- Designed streaming architectures (WebSocket, SSE) for real-time AI-powered features with <100ms latency, powering live dashboards for fintech and healthcare clients
- Built custom data pipelines processing 2M+ daily events for ML feature stores, enabling real-time model inference across 3 production environments